Similarly Cerium and terbium attain $4f^0$ and $4f^7$ configurations respectively in the +4 oxidation states. Eu^{2+} and Yb^{2+} ions have exactly half filled and completely filled f orbitals respectively.

The stability of different oxidation states has an impact on the properties of these elements. the following table shows the different oxidation states of lanthanoids.

Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
		+2		+2	+2						+2	+2	
+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3
+4	+4	+4					+4	+4					

Atomic and ionic radii:

As we move across 4f series, the atomic and ionic radii of lanthanoids show gradual decrease with increse in atomic number. This decrese in ionic size is called lanthanoid contraction.

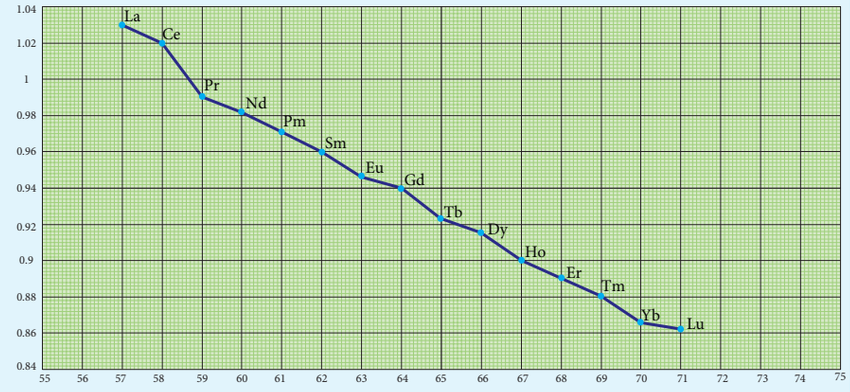


Figure 4.11 Variation of atomic radii of lanthanoids

Cause of lanthanoid contraction:

As we move from one element to another in 4f series (Ce to Lu) the nuclear charge increases by one unit and an additional electron is added into the same inner 4f sub shell. We know that 4f sub shell have a diffused shapes and therefore the shielding effect of 4f elelctrons relatively poor.hence, with increase of nuclear charge, the valence shell is pulled slightly towards nucleus. As a result, the effetive nuclear charge experienced by the 4f elelctorns increases and the size of Ln^{3+} ions decreases. Lanthanoid contraction of various lanthanoids is shown in the graph

Consequences of lanthanoid contraction:

1. Basic nature

As we from Ce^{3+} to Lu^{3+} , the basic character of Ln^{3+} ions decrease. Due to the decrease in the size of Ln^{3+} ions, the ionic character of Ln–OH bond decreases (covalent character increases) which results in the decrease in the basic nature.

2. Similarities among lanthanoids:

In the complete f - series only 10 pm decrease in atomic radii and 20 pm decrease in ionic radii is observed. because of this very small change in radii of lanthanoids, their chemical properties are quite similar.

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